# Shiva Voggu

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## EDUCATION

| Institution | Degree / Board | Passing Year | Score / CGPA |
|---|---|---|---|
| Indian Institute of Technology Hyderabad | B.Tech | 2027 | 8.44 |
| NRUPATUNGA JUNIOR COLLEGE | Class XII (TELANGANA STATE BOARD OF INTERMEDIATE EDUCATION) | 2018 | 84.00% |
| M J P T BC W RESIDENTIAL SCHOOL | Class X (TELANGANA STATE SCHOOL SECONDARY BOARD) | 2016 | 9.20% |

## WORK EXPERIENCE

**FEA Engineer** (**April 2023 - May 2025)**

**Advanced Systems Laboratory(ASL), DRDO, Hyderabad**

- Contributed to **transient thermal simulation** of the **Hypersonic Glide Vehicle (HGV)** -- one of India's most advanced missile programmes (**range: 6000-10000 km**) -- analysing thermal behaviour under extreme hypersonic (**Mach 5+**) loading conditions
- Executed **transient thermal analysis** on **missile structures** using **ANSYS APDL** and **ANSYS Workbench**, ensuring thermal performance under critical defence-grade loading conditions
- Developed and optimized **complex FE models** primarily using **ANSYS APDL** for **pre-processing** and **meshing of complex geometries**, with selective use of **HyperMesh** for specific cases
- Gained in-depth understanding of **advanced materials** used in **missile structure design** including **Titanium Alloys**, **Carbon Composites (C-SiC tiles)**, **Carbon-Carbon 3D**, **Carbon-Carbon 4D**, **Si-Si Foam**, **Carbon Foam**, and **Inconel 718**, and their thermal behaviour under hypersonic conditions
- Witnessed **real-time thermal testing** of the HGV and **contributed** transient thermal analysis to **LRAShM** -- one of India's most successful recent missile programmes and **SRBM (Short Range Ballistic Missile)** structural components, developing practical understanding of flight trajectory and aerodynamic behaviour at hypersonic speeds

**Internships (April 2021 - May 2021)**

- **Vehicle dynamics** -- Analysed vehicle motion and response to driver inputs, propulsion system outputs, and ambient conditions such as air, surface, and water
- **NDT Workshop** -- Attended hands-on workshop on **Non-Destructive Testing (NDT) techniques**, gaining practical understanding of inspection methodologies used in engineering applications

## PROJECTS

**MTech Project:**

**Experimental Characterisation and FEM Validation of Johnson-Cook Model of Steel for High-Speed Projectile Impact.**

- **Characterised yield** and **fracture behaviour of steel** using **Uniaxial Tension Testing** and **Digital Image Correlation** across axisymmetric and plane strain notched specimens at varying stress triaxiality and Lode parameter combinations
- Determined **Johnson-Cook plasticity** and **failure parameters** from full-field strain measurements, capturing **yield stress, strain hardening** and **strain-rate sensitivity coefficients**
- **Validated** the obtained constitutive model parameters against FEM simulations to ensure predictive accuracy under multi-axial stress states

**Visco-hyperelastic constitutive modeling of strain rate sensitive soft materials validation**

- **Investigated visco-hyperelastic behaviour of strain-rate-sensitive soft materials**, modelling large deformation, **nonlinear stress-strain response**, and time-dependent viscous dissipation using an **external-state-variable-driven**

**constitutive framework**.
- **Calibrated constitutive parameters from experimental stress-stretch data** across different deformation/strain-rate conditions to capture both hyperelastic and viscous material responses.

**Characterisation of Materials Under High Strain Rate Using Split Hopkinson Pressure Bar (SHPB)**

- **Characterising dynamic mechanical behaviour** of materials under high strain rates using Split Hopkinson Pressure Bar (SHPB) technique, analysing stress-strain response and rate-dependent material properties under **impact loading conditions.**

**FEA-Based Buckling and Failure Analysis of a Stiffened Composite Aircraft Wing Skin Panel Using Double-Double Laminate Design**

- Performed composite **structural FEA** using **ANSYS Workbench** and **ANSYS ACP** to investigate pre-buckling, buckling, and post-buckling behavior of a **stiffened CFRP aircraft wing-skin panel** under compressive loading.
- Transitioned the baseline **conventional laminate** to a **Double-Double (DD) laminate** architecture and evaluated its effect on **structural stability**, buckling resistance, load-carrying capability, and lightweight composite design.
- Applied **Hashin failure criteria** to assess fiber/matrix failure modes and evaluated the resulting failure responses, obtaining physically consistent trends with published composite-structure research.

**B.E. Project: Characterisation of Resistance Spot Weldments**

- Investigated mechanical and metallurgical properties of resistance spot weld joints
- Analysed weld nugget formation, tensile shear strength and microstructural behaviour

## SKILLS

- **Simulation & CAE Tools** -- ANSYS APDL | Abaqus | ANSYS Workbench | ANSYS Discovery | HyperMesh | ANSYS ACP
- **CAD & Design** -- Ansys Spaceclaim | CATIA | SolidWorks | AutoCAD
- **Programming & Computing** -- MATLAB | Python | Microsoft Office Suite
- **Engineering Expertise** -- Finite Element Analysis (FEA) | Thermal & Structural Analysis | Digital Image Correlation (DIC)

## RELEVANT COURSES

Design& Analysis of Composite Structures | Strength of Materials | Fracture Mechanics | Vibrations & Dynamics | Machine Learning | FEM | Advanced Mechanics of Solids |

## POSITIONS OF RESPONSIBILITY

**Mentor & Technical Guide** -- Advanced Systems Laboratory (ASL), DRDO, Hyderabad

- **Mentored** and guided junior colleagues on FEA workflows, transient thermal and structural analysis procedures, and simulation tools including ANSYS and HyperMesh over one year, significantly enhancing team efficiency and technical capability

## EXTRACURRICULAR ACTIVITIES & INTERESTS

- **Swing Trader(Investment Planner)** -- Actively trade equities on Indian stock exchanges(NSE/BSE), developing strong analytical thinking, risk assessment, and quick decision-making skills
- **Sports & Fitness** -- Regular gym training, active sports participation, and maintaining a healthy diet, complemented by reading self-help books focused on personal growth and continuous self-improvement